

A SHARP LOGARITHMIC BOUND FOR ENTROPY LOSS UNDER ADDITION

GPT-6 ASTRA

ABSTRACT. We sharpen an entropy inequality from [KK26] by replacing its linear dependence on the number of summands by an optimal logarithmic factor. The entropy lost under addition of independent finitely supported real random variables is bounded by 62 times the total variance energy multiplied by the entropy of the normalized individual energies. We also improve the corresponding estimate for variance energy below a prescribed scale.

This is a unedited model output on the basis of the paper [KK26] with the prompting done by C.K. and using the prompters local context.

1. INTRODUCTION

Let $X_1, \dots, X_m$ be independent finitely supported real random variables. We write

$$\mathcal{L}(X_1, \dots, X_m) = \sum_{i=1}^m H(X_i) - H\left(\sum_{i=1}^m X_i\right) = H\left(X_1, \dots, X_m \left| \sum_{i=1}^m X_i \right.\right),$$

where H denotes Shannon entropy and all logarithms are natural. Theorem 1.3 of [KK26] states that

$$\mathcal{L}(X_1, \dots, X_m) \leq 30m \sum_{i=1}^m \mathcal{E}(X_i). \quad (1.1)$$

Here $\mathcal{E}$ is the variance energy, which measures local variance averaged over all scales. The discussion following that theorem notes that the factor m is unlikely to be optimal. We determine its optimal order and prove a refinement which takes account of unequal individual energies.

We use precisely the normalization of [KK26]. For a finitely supported probability measure ρ on $\mathbb{R}$, put $I(a, r) = [a, a + r)$ and

$$F_\rho(I) = \inf_{c \in \mathbb{R}} \int_I (x - c)^2 d\rho(x) = \rho(I) \operatorname{Var}(\rho|_I), \quad (1.2)$$

$$V_\rho(r) = \frac{4}{r^3} \int_{\mathbb{R}} F_\rho(I(a, r)) da, \quad (1.3)$$

$$\mathcal{E}_R(\rho) = \int_0^R V_\rho(r) \frac{dr}{r}, \quad \mathcal{E}(\rho) = \mathcal{E}_\infty(\rho). \quad (1.4)$$

The restriction $\rho|_I$ in (1.2) is normalized to be a probability measure; a term with $\rho(I) = 0$ is defined to be zero. We write $\mathcal{E}(X)$ for the energy of the law of X .

For $a = (a_1, \dots, a_m) \in [0, \infty)^m$, put $A = \sum_i a_i$ and define

$$\mathfrak{H}(a) = \sum_{i=1}^m a_i \log \frac{A}{a_i} = A \log A - \sum_{i=1}^m a_i \log a_i. \quad (1.5)$$

We use $0 \log 0 = 0$, so in particular $\mathfrak{H}(0) = 0$. If $A > 0$, then $\mathfrak{H}(a) = AH(a/A)$.

Theorem 1.1. *Let $m \geq 1$ and let $X_1, \dots, X_m$ be independent finitely supported real random variables. Then*

$$\mathcal{L}(X_1, \dots, X_m) \leq 62 \mathfrak{H}(\mathcal{E}(X_1), \dots, \mathcal{E}(X_m)). \quad (1.6)$$

Consequently, for $m \geq 2$,

$$\mathcal{L}(X_1, \dots, X_m) \leq 62 \log m \sum_{i=1}^m \mathcal{E}(X_i). \quad (1.7)$$

The logarithmic factor in (1.7) is optimal. To make this precise, let

$$C_m = \sup \frac{\mathcal{L}(X_1, \dots, X_m)}{\sum_i \mathcal{E}(X_i)},$$

where the supremum is over the variables in Theorem 1.1 with positive denominator.

Theorem 1.2. *For every $m \geq 2$,*

$$\frac{3}{2} \log m \leq C_m \leq 62 \log m. \quad (1.8)$$

More generally, let $p = (p_1, \dots, p_m)$ be a probability vector with $H(p) > 0$, and let $B_i^{(\varepsilon)}$ be independent Bernoulli variables with parameters εp_i . Then

$$\lim_{\varepsilon \downarrow 0} \frac{\mathcal{L}(B_1^{(\varepsilon)}, \dots, B_m^{(\varepsilon)})}{\mathfrak{H}(\mathcal{E}(B_1^{(\varepsilon)}), \dots, \mathcal{E}(B_m^{(\varepsilon)}))} = \frac{3}{2}. \quad (1.9)$$

Thus the dependence on the normalized energy vector in (1.6) is sharp up to an absolute constant.

The proof actually gives the coefficient $36 + 18/\log 2 < 62$ in both (1.6) and (1.7). We have not tried to optimize this constant. Example 4.2, using two independent uniform laws, shows that any universal coefficient in either inequality must be at least $3/(2 \log 2) = 2.164\dots$. We also do not determine the leading constant in C_m as m tends to infinity.

Since variance energy is invariant under nonconstant affine changes of variables, the same results apply to $\sum_i a_i X_i$ with arbitrary nonzero real coefficients a_i . Coordinates with zero coefficient are simply omitted from the corresponding entropy loss.

Proof outline and relation to previous work. The proof retains the Poisson partitions of [KK26, Section 2]. Write

$$D(X) = \mathbb{E}|X - X'|, \quad A(X | S) = \int_{\mathbb{R}} H(\mathbf{1}_{\{X \leq a\}} | S) da,$$

where X' is an independent copy of X . The unconditional functional $A(X)$ is the cumulative paired Shannon entropy studied in [KMD16]. The conditional functional and the prediction rule used below are those of [KK26].

The estimate [KK26, Lemma 2.1] is $A(X \mid X + Y) \leq 9(D(X) + D(Y))$ for independent X, Y . We strengthen its form to

$$A(X \mid X + Y) \leq C_0 \Phi(D(X), D(Y)), \quad \Phi(a, b) = a \log \frac{a+b}{a} + b \log \frac{a+b}{b}. \quad (1.10)$$

In particular, if $0 < a \leq b$, the right-hand side is of order $a \log(1 + b/a)$ rather than $a + b$. This distinction is essential when one variable is compared with the sum of the other $m - 1$ variables. An elementary entropy inequality then sums (1.10) without a linear loss in m . The Poisson dispersion estimate of [KK26, Lemma 2.2] converts the resulting bound to variance energy. We include the required partition and energy arguments so that the proof is self-contained. Section 5 gives the resulting improvement of [KK26, Proposition 3.3].

2. A REFINED CONDITIONAL CUMULATIVE ENTROPY ESTIMATE

We extend Φ continuously to $[0, \infty)^2$, with $\Phi(a, 0) = \Phi(0, a) = 0$. It is concave, homogeneous of degree one, and nondecreasing in each coordinate. These assertions follow either from $\Phi(a, b) = (a + b)h_2(a/(a + b))$, where

$$h_2(u) = -u \log u - (1 - u) \log(1 - u),$$

or by differentiation. In particular, for nonnegative integrable functions f, g ,

$$\int \Phi(f, g) \leq \Phi\left(\int f, \int g\right). \quad (2.1)$$

For a finite measure this is Jensen's inequality and homogeneity; the general case follows by restriction and monotone convergence. We shall use the same observation for $\mathfrak{H}$.

Lemma 2.1. *Let Z, W be independent random variables taking values in $(0, 1]$, each stochastically dominating a uniform random variable on $(0, 1)$. Put $a = \mathbb{E}(1 - Z)$ and $b = \mathbb{E}(1 - W)$. Then*

$$\mathbb{E} \log \left(1 + \frac{1 - Z}{Z} \frac{1 - W}{W} \right) \leq C_0 \Phi(a, b) \quad (2.2)$$

with $C_0 = 6 + 3/\log 2$.

Proof. Write $R = (1 - Z)/Z$ and $T = (1 - W)/W$. We have $a, b \leq 1/2$. If either is zero there is nothing to prove, so assume both are positive. Let $p = \mathbb{P}(Z < 1/2) \leq 2a$ and $q = \mathbb{P}(W < 1/2) \leq 2b$. The quantile function of a variable stochastically dominating uniform is at least the identity. Consequently, for every $s \geq 0$,

$$\mathbb{E}[\mathbf{1}_{\{Z < 1/2\}} \log(1 + s/Z)] \leq \int_0^p \log(1 + s/u) du = \Phi(p, s). \quad (2.3)$$

Indeed, the event that the quantile is below $1/2$ corresponds, up to a null set, to the initial interval of length p .

On $\{Z \geq 1/2, W \geq 1/2\}$, use $\log(1 + RT) \leq RT$, $R \leq 2(1 - Z)$ and $T \leq 2(1 - W)$. Independence bounds this contribution by $4ab$. On $\{Z < 1/2, W \geq 1/2\}$, apply (2.3), then concavity in the second coordinate and monotonicity, to obtain

$$\begin{aligned} \mathbb{E}[\mathbf{1}_{\{Z < 1/2, W \geq 1/2\}} \log(1 + RT)] &\leq \mathbb{E}\Phi(p, T\mathbf{1}_{\{W \geq 1/2\}}) \\ &\leq \Phi(2a, 2b) = 2\Phi(a, b). \end{aligned}$$

The other mixed region gives the same bound.

For the remaining region, use independent quantile representations and $R \leq 1/Z$, $T \leq 1/W$. Its contribution is at most

$$\begin{aligned} \int_0^p \int_0^q \log(1 + 1/(uv)) \, dv \, du &\leq pq \left(\log \frac{2}{pq} + 2 \right) \\ &\leq 4ab \left(\log \frac{1}{ab} + 2 \right). \end{aligned}$$

Here $u, v \leq 1$ in the first inequality, and the function $s \mapsto s(\log(2/s) + 2)$ is increasing for $0 < s \leq 1$ in the second. Combining the four regions, the left-hand side of (2.2) is bounded by

$$4\Phi(a, b) + 4ab \left(\log \frac{1}{ab} + 3 \right).$$

To track the constant, suppose by symmetry that $a \leq b \leq 1/2$, and put $r = b/a \geq 1$ and $\psi(r) = \Phi(a, b)/a = \log(1+r) + r \log(1+1/r)$. Since $\psi(1) = 2 \log 2$ and $r\psi'(r) = r \log(1+1/r) \geq \log 2$, we have $\psi(r) \geq (2 + \log r) \log 2$. Also, $b(\log r + 2 \log(1/b) + 3)$ is increasing for $0 < b \leq 1/2$. Consequently,

$$\frac{ab(\log(1/(ab)) + 3)}{\Phi(a, b)} \leq \frac{\frac{1}{2} \log r + \log 2 + \frac{3}{2}}{(2 + \log r) \log 2} \leq \frac{1}{2} + \frac{3}{4 \log 2}.$$

The preceding bound therefore proves the lemma with $C_0 = 4 + 4(\frac{1}{2} + \frac{3}{4 \log 2}) = 6 + 3/\log 2$. $\square$

Proposition 2.2. *For independent finitely supported real random variables X, Y ,*

$$A(X \mid X + Y) \leq C_0 \Phi(D(X), D(Y)), \quad (2.4)$$

where C_0 may be taken to be the constant in Lemma 2.1.

Proof. We use the prediction rule in the proof of [KK26, Lemma 2.1], keeping the logarithmic loss instead of estimating it by a square root. Let F_X, F_Y be the distribution functions, and for $u > 0$ put

$$T_X(u) = \mathbb{P}(X' > X + u), \quad T_Y(u) = \mathbb{P}(Y' > Y + u).$$

Since $F_X(X)$ stochastically dominates uniform, so does $F_X(X+u)$, and its expected complement is $T_X(u)$. Similarly, $1 - F_Y(Y - u)$ stochastically dominates uniform. Its expected complement is $\mathbb{P}(Y' \leq Y - u) = T_Y(u)$ for almost every $u > 0$; only the finitely many positive atom differences can be exceptions.

Given $S = X + Y = s$, predict $\{X \leq v\}$ with probability

$$q_s(v) = \frac{F_X(v)(1 - F_Y(s - v))}{F_X(v)(1 - F_Y(s - v)) + (1 - F_X(v))F_Y(s - v)}.$$

Values at zero denominators may be chosen arbitrarily. For a realized pair (x, y) and $v \neq x$, the predicted probability of the realized answer is positive. The binary relative entropy inequality gives

$$H(\mathbf{1}_{\{X \leq v\}} \mid S) \leq \mathbb{E}[-\mathbf{1}_{\{X \leq v\}} \log q_S(v) - \mathbf{1}_{\{X > v\}} \log(1 - q_S(v))]. \quad (2.5)$$

Integrate in v and use Tonelli. On $v = x + u$, $u > 0$, the loss is

$$\log \left(1 + \frac{1 - F_X(x + u)}{F_X(x + u)} \frac{F_Y(y - u)}{1 - F_Y(y - u)} \right).$$

By Lemma 2.1, its expectation is at most $C_0\Phi(T_X(u), T_Y(u))$. On $v = x - u$ the loss is

$$\log \left(1 + \frac{F_X(x-u)}{1-F_X(x-u)} \frac{1-F_Y(y+u)}{F_Y(y+u)} \right),$$

and the reflected application of the same lemma gives the same bound. Finally,

$$\int_0^\infty T_X(u) du = \frac{1}{2}D(X), \quad \int_0^\infty T_Y(u) du = \frac{1}{2}D(Y).$$

Thus (2.1) and homogeneity give

$$A(X | S) \leq 2C_0 \int_0^\infty \Phi(T_X(u), T_Y(u)) du \leq C_0\Phi(D(X), D(Y)). \quad \square$$

Lemma 2.3. *The function $\mathfrak{H}$ is concave, homogeneous of degree one, and nondecreasing in each coordinate. If $a_i \geq 0$ and $A = \sum_i a_i$, then*

$$\sum_{i=1}^m \Phi(a_i, A - a_i) \leq 2\mathfrak{H}(a) \leq 2A \log m. \quad (2.6)$$

Proof. Concavity follows from the perspective representation $\mathfrak{H}(a) = AH(a/A)$. In the interior, $\partial\mathfrak{H}/\partial a_i = \log(A/a_i) \geq 0$; the assertion on the boundary follows by continuity. Homogeneity is immediate.

For the first inequality in (2.6), normalize $A = 1$ and write $p_i = a_i$. It suffices to show

$$-\sum_i (1-p_i) \log(1-p_i) \leq H(p). \quad (2.7)$$

This is the entropy-extropy inequality of [LSA15, Result 1]. The following elementary proof is useful here. Since $\prod_{i \neq j} (1-p_i) \geq 1 - \sum_{i \neq j} p_i = p_j$, we have

$$-\sum_i (1-p_i) \log(1-p_i) = \sum_j p_j \sum_{i \neq j} -\log(1-p_i) \leq \sum_j -p_j \log p_j.$$

Boundary cases follow by continuity. Now $\sum_i \Phi(p_i, 1-p_i) = \sum_i h_2(p_i) \leq 2H(p)$, and $H(p) \leq \log m$ proves the last inequality. $\square$

3. POISSON PARTITIONS AND THE ENTROPY BOUND

We first record the energy facts needed below. These are [KK26, Lemmas 2.2 and 2.3]; we give the proofs with our notation.

Lemma 3.1. *The functionals $F_\rho(I)$, $V_\rho(r)$ and $\mathcal{E}_R(\rho)$ are concave in ρ . For $a \neq 0$,*

$$V_{aX+b}(r) = V_X(r/|a|), \quad \mathcal{E}(aX+b) = \mathcal{E}(X).$$

For a finitely supported law of diameter at most D , $\mathcal{E}(\rho) < \infty$ and

$$\mathcal{E}(\rho) - \mathcal{E}_R(\rho) \leq \frac{D^2}{2R^2}. \quad (3.1)$$

Proof. The infimum in (1.2) is an infimum of linear functions of ρ , which proves concavity; integration preserves it. A change of variables gives the affine identities. The conditional variance is at most $D^2/4$, and $\int_{\mathbb{R}} \rho(I(a, r)) da = r$, so $V_\rho(r) \leq D^2/r^2$. Integration proves (3.1). For finite support, $V_\rho(r) = 0$ below the smallest positive atom spacing. A point mass has zero energy. These observations also prove finiteness. $\square$

Place Poisson points on $\mathbb{R} \times (0, \infty)$ with intensity $da dt$. At time $t > 0$, the spatial coordinates of points with time coordinate at most t cut $\mathbb{R}$ into a partition $\mathcal{Q}_t$. All input variables are independent of the cuts. Conditional expressions involving $\mathcal{Q}_t X$ always condition on the partition as well as the cell containing X . Write

$$d_X(t) = \mathbb{E}_{\mathcal{Q}} D(X \mid \mathcal{Q}_t X), \quad v_X(t) = \mathbb{E}_{\mathcal{Q}} \text{Var}(X \mid \mathcal{Q}_t X),$$

where the expectation includes the cell probabilities.

Lemma 3.2 (Poisson dispersion). *For every finitely supported real random variable X ,*

$$\int_0^\infty d_X(t) dt \leq 2 \int_0^\infty t v_X(t) dt = 3\mathcal{E}(X). \quad (3.2)$$

Proof. Put $\rho = \mathcal{L}(X)$. At intensity t , cells with left endpoint a and length r have intensity $t^2 e^{-tr} da dr$. Hence

$$v_X(t) = \int_0^\infty \int_{\mathbb{R}} t^2 e^{-tr} F_\rho(I(a, r)) da dr.$$

Tonelli's theorem and $\int_0^\infty t^3 e^{-tr} dt = 6r^{-4}$ yield

$$\int_0^\infty t v_X(t) dt = 6 \int_0^\infty \int_{\mathbb{R}} \frac{F_\rho(I(a, r))}{r^4} da dr = \frac{3}{2} \mathcal{E}(X). \quad (3.3)$$

For independent copies X, X' , conditioning on the partition gives

$$\mathbb{E}[(X - X')^2 \mathbf{1}_{\{\mathcal{Q}_t X = \mathcal{Q}_t X'\}} \mid \mathcal{Q}_t] = 2 \sum_{I \in \mathcal{Q}_t} \rho(I)^2 \text{Var}(\rho|_I) \leq 2 \sum_{I \in \mathcal{Q}_t} \rho(I) \text{Var}(\rho|_I).$$

Two points at distance s belong to the same cell with probability e^{-ts} . Averaging and integrating therefore show that

$$D(X) \leq 2 \int_0^\infty v_X(t) dt.$$

Apply this inequality to the conditional law in each cell at time t . Independent cuts of intensity u refine the partition to one with the law of $\mathcal{Q}_{t+u}$, so

$$d_X(t) \leq 2 \int_t^\infty v_X(s) ds.$$

Integration in t , followed by (3.3), proves (3.2). $\square$

Proof of Theorem 1.1. For each input X_i , take an independent Poisson partition $\mathcal{Q}_t^{(i)}$, and put $S = \sum_i X_i$. As in [KK26, proof of Theorem 1.3], consider

$$J(t) = \mathbb{E}_{\mathcal{Q}} H(\mathcal{Q}_t^{(1)} X_1, \dots, \mathcal{Q}_t^{(m)} X_m \mid S).$$

At time zero there are no cuts, so $J(0) = 0$. Almost surely every gap between consecutive atoms eventually receives a cut. The cell labels therefore eventually determine every input, and bounded convergence gives

$$J(\infty) = H(X_1, \dots, X_m \mid S) = \mathcal{L}(X_1, \dots, X_m). \quad (3.4)$$

For fixed cuts, let $B = B_1 \times \dots \times B_m$ range over products of cells, and write $X = (X_1, \dots, X_m)$. A new cut in coordinate i at a adds the conditional entropy of $\mathbf{1}_{\{X_i \leq a\}}$ given S and the previous labels. Consequently,

$$J'(t) = \mathbb{E}_{\mathcal{Q}} \sum_B \mathbb{P}(X \in B) \sum_{i=1}^m A(X_i \mid S, X \in B). \quad (3.5)$$

Terms of zero probability are omitted. Here $A(X_i | S, X \in B)$ is computed under the conditional law given $X \in B$. To justify the identity, restrict cuts to fixed compact intervals containing the supports. The process then has finite rate; the probability of two or more new cuts in a time interval of length h is $O(h^2)$. The entropy increment from one cut is given by the chain rule, and all entropies are bounded. Dividing by h and letting h decrease to zero proves (3.5). Cuts outside these intervals do not affect the information in the labels.

Under $X \in B$, the coordinates remain independent. Put

$$a_i(B) = D(X_i | X_i \in B_i), \quad A(B) = \sum_i a_i(B).$$

The triangle inequality, with independent copies of all coordinates, gives

$$D\left(\sum_{j \neq i} X_j \middle| X \in B\right) \leq \sum_{j \neq i} a_j(B).$$

By Proposition 2.2, monotonicity of Φ and Lemma 2.3,

$$\sum_i A(X_i | S, X \in B) \leq C_0 \sum_i \Phi(a_i(B), A(B) - a_i(B)) \leq 2C_0 \mathfrak{H}(a(B)).$$

Averaging over the cells and cuts, concavity of $\mathfrak{H}$ yields

$$J'(t) \leq 2C_0 \mathfrak{H}(d_1(t), \dots, d_m(t)), \quad d_i(t) = d_{X_i}(t). \quad (3.6)$$

Let $e_i = \mathcal{E}(X_i)$ and $b_i = \int_0^\infty d_i(t) dt$. Lemma 3.2 gives $b_i \leq 3e_i$. Integrate (3.6), using Jensen's inequality and homogeneity on each finite time interval and then passing to the limit. By monotonicity,

$$J(\infty) \leq 2C_0 \mathfrak{H}(b_1, \dots, b_m) \leq 6C_0 \mathfrak{H}(e_1, \dots, e_m).$$

Since $6C_0 = 36 + 18/\log 2 < 62$, this and (3.4) prove (1.6). The normalized energy vector has entropy at most $\log m$, which gives (1.7). The zero-energy case is understood by continuity. $\square$

4. SHARPNESS

Lemma 4.1. *If B is Bernoulli with parameter p , then*

$$\mathcal{E}(B) = \frac{2}{3}p(1-p). \quad (4.1)$$

Proof. For $r \leq 1$, an interval of length r contains at most one atom, apart from irrelevant endpoints, so $V_B(r) = 0$. For $r > 1$, the set of left endpoints for which both atoms lie in the interval has length $r - 1$. On such an interval $F = p(1-p)$. Thus

$$\mathcal{E}(B) = 4p(1-p) \int_1^\infty \frac{r-1}{r^4} dr = \frac{2}{3}p(1-p). \quad \square$$

Proof of Theorem 1.2. Fix a probability vector p with $H(p) > 0$, and put $N_\varepsilon = \sum_i B_i^{(\varepsilon)}$. As $\varepsilon \downarrow 0$, with m and p fixed,

$$\mathbb{P}(N_\varepsilon = 1) = \varepsilon + O(\varepsilon^2), \quad \mathbb{P}(N_\varepsilon \geq 2) = O(\varepsilon^2).$$

Conditionally on $N_\varepsilon = 1$, the index of the unique nonzero coordinate has probabilities

$$\frac{\varepsilon p_i \prod_{j \neq i} (1 - \varepsilon p_j)}{\mathbb{P}(N_\varepsilon = 1)} = p_i + O(\varepsilon).$$

Given $N_\varepsilon = n$, the vector of Bernoulli variables has entropy at most $m \log 2$. Continuity of entropy on a finite simplex therefore gives

$$\mathcal{L}(B_1^{(\varepsilon)}, \dots, B_m^{(\varepsilon)}) = \varepsilon H(p) + o(\varepsilon). \quad (4.2)$$

On the other hand, Lemma 4.1 gives

$$e_i(\varepsilon) = \frac{2}{3}\varepsilon p_i(1 - \varepsilon p_i), \quad \sum_i e_i(\varepsilon) = \frac{2}{3}\varepsilon + O(\varepsilon^2).$$

The normalized energy vector tends to p . Hence

$$\mathfrak{H}(e(\varepsilon)) = \frac{2}{3}\varepsilon H(p) + o(\varepsilon),$$

which proves (1.9). Taking $p_i = 1/m$ in (4.2) shows that the ratio of the loss to the total energy tends to $\frac{3}{2} \log m$. This proves the lower bound in (1.8); the upper bound is Theorem 1.1. $\square$

Example 4.2 (A lower bound for the absolute constant). *Let U_N, U'_N be independent and uniform on $\{1, \dots, N\}$. Then*

$$\mathcal{E}(U_N) = \frac{1}{3} \log N + O(1). \quad (4.3)$$

Indeed, for $1 \leq r \leq N/2$, intervals of length r contained in $[1, N]$ contain $k = r + O(1)$ consecutive integers, whose uniform law has variance $(k^2 - 1)/12$. Their left endpoints form an interval of length $N + O(r)$, so their contribution to $V_{U_N}(r)$ is $1/3 + O(1/r + r/N)$. Intervals meeting either endpoint have left endpoints in a set of length $O(r)$, and contribute $O(r/N)$. Thus

$$V_{U_N}(r) = \frac{1}{3} + O\left(\frac{1}{r} + \frac{r}{N}\right) \quad (1 \leq r \leq N/2).$$

Below scale 1 the local variance vanishes, and above scale $N/2$ the energy is $O(1)$ by (3.1). Integration proves (4.3). Moreover,

$$\log N \leq H(U_N + U'_N) \leq \log(2N - 1),$$

where the lower bound follows by conditioning on U'_N . Hence $\mathcal{L}(U_N, U'_N) = \log N + O(1)$ and

$$\lim_{N \rightarrow \infty} \frac{\mathcal{L}(U_N, U'_N)}{2\mathcal{E}(U_N) \log 2} = \frac{3}{2 \log 2}.$$

The denominator is both $\log 2 (\mathcal{E}(U_N) + \mathcal{E}(U'_N))$ and $\mathfrak{H}(\mathcal{E}(U_N), \mathcal{E}(U'_N))$. This proves the claimed lower bound for the constant in either inequality.

5. VARIANCE ENERGY BELOW A PRESCRIBED SCALE

For a finitely supported real random variable Y , use the entropy at scale $R > 0$ from [Var19, Sections 1.2 and 2.2]:

$$H(Y; R) = \frac{1}{R} \int_0^R H\left(\left\lfloor \frac{Y+t}{R} \right\rfloor\right) dt. \quad (5.1)$$

We obtain the following sharpened form of [KK26, Proposition 3.3].

Proposition 5.1. *Suppose $Y = \sum_{j=1}^L a_j D_j$, where $a_j \in \mathbb{R}$ and the D_j are independent real random variables, each taking at most K values. For every $R > 0$ and integer $m \geq 2$,*

$$\mathcal{E}_R(Y) \geq \frac{1}{62 \log m} \left(H(Y) - H(Y; R) - \frac{L}{m} \log \binom{m+K-1}{K-1} \right) - \frac{1}{2}, \quad (5.2)$$

Proof. Take independent copies $Y_1, \dots, Y_m$ of the entire digit sum, and put $S = \sum_i Y_i$. For each digit position, the sum of its m copies is determined by the multiplicities of its at most K possible values. There are at most $\binom{m+K-1}{K-1}$ multiplicity vectors, so

$$H(S) \leq L \log \binom{m+K-1}{K-1}. \quad (5.3)$$

Fix $t \in [0, R)$ and set $Q_i = \lfloor (Y_i + t)/R \rfloor$, with Q defined analogously for Y . The variables Y_i remain independent conditionally on $(Q_1, \dots, Q_m)$. Apply (1.7) to these conditional laws and average. Since conditioning decreases entropy,

$$mH(Y | Q) \leq H(S) + 62m \log m \mathbb{E} \mathcal{E}(\mathcal{L}(Y | Q)).$$

Each conditional law is supported on an interval of length R . Lemma 3.1 therefore gives

$$\mathbb{E} \mathcal{E}(\mathcal{L}(Y | Q)) \leq \mathbb{E} \mathcal{E}_R(\mathcal{L}(Y | Q)) + \frac{1}{2} \leq \mathcal{E}_R(Y) + \frac{1}{2}.$$

Finally, the entropy chain rule implies

$$\frac{1}{R} \int_0^R H \left(Y \middle| \left\lfloor \frac{Y+t}{R} \right\rfloor \right) dt = H(Y) - H(Y; R).$$

Average in t , use (5.3), and divide by m to obtain (5.2). $\square$

The elementary estimate $\binom{m+K-1}{K-1} \leq (m+1)^{K-1}$ recovers the error term of [KK26, Proposition 3.3], while replacing its denominator $30m$ by $62 \log m$. One may also eliminate the auxiliary parameter m .

Corollary 5.2. *Under the hypotheses of Proposition 5.1, suppose $K \geq 2$ and $\Delta = H(Y) - H(Y; R) > 0$. Put $q = L(K-1)$. There is an absolute constant $c > 0$ such that*

$$\mathcal{E}_R(Y) + \frac{1}{2} \geq \frac{c\Delta}{\log(2 + q/\Delta)}. \quad (5.4)$$

Proof. We have $\Delta \leq H(Y) \leq L \log K \leq q$, so $a = q/\Delta \geq 1$. Choose $m = \lceil Ma \log(2 + a) \rceil$, where M is a sufficiently large absolute constant. Then

$$\frac{q \log(m+1)}{m} \leq \frac{\Delta}{2}, \quad \log m \leq C_1 \log(2 + a)$$

for an absolute constant C_1 . Substitution in (5.2), using the multiplicity bound above, proves (5.4). $\square$

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